

Half a Century of Stereotyping Associations Between Gender and Intellectual Ability in Films

Ramiro H. Gálvez

Universidad de Buenos Aires and Instituto de Ciencias de la Computación, CONICET-UBA

Valeria Tiffenberg

Universidad de Buenos Aires

Edgar Altszyler

Universidad de Buenos Aires, Instituto de Ciencias de la Computación, CONICET-UBA, and

Fundación Sadosky

Author Note

Ramiro H. Gálvez, Departamento de Computación, FCEyN, Universidad de Buenos Aires and Instituto de Ciencias de la Computación, CONICET-UBA; Valeria Tiffenberg, Departamento de Computación, FCEyN, Universidad de Buenos Aires; Edgar Altszyler, Departamento de Computación, FCEyN, Universidad de Buenos Aires, Instituto de Ciencias de la Computación, CONICET-UBA, and Fundación Sadosky

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Correspondence concerning this manuscript should be addressed to Ramiro H. Gálvez, Departamento de Computación, FCEyN, Universidad de Buenos Aires and Instituto de Ciencias de la Computación, CONICET-UBA. Email: rgalvez@dc.uba.ar

Abstract

A particularly longstanding, prevalent, and well-documented stereotype is the belief that men possess higher-level cognitive abilities than women do. This brilliance = male stereotype has been shown to be endorsed even by children as young as 6-years-old and is believed to be a factor driving the underrepresentation of women in STEM fields. Motivated by the fact that cultural products serve as a source for acquiring individual values and behaviors, we study the presence of this stereotype in a large collection of movie transcripts covering half a century of Western-world film history ($n = 11,550$). Concretely, we use natural language processing techniques to quantify associations between gender pronouns and high-level cognitive ability-related words. Overall, our estimates suggest that, at an aggregate level, the brilliance = male stereotype is effectively present in films and that movies specifically targeted at children contain this stereotypical association. Moreover, this pattern seems to have been quite persistent for the last 50 years.

Keywords: gender stereotypes, brilliance = male stereotype, STEM fields, film history, culturomics, computational content analysis

Half a Century of Stereotyping Associations Between Gender and Intellectual Ability in Films

A particularly longstanding, prevalent and well-documented stereotype is the belief that men possess higher-level cognitive abilities than women do (Broverman 1970; Kirkcaldy et al. 2007; Upson & Friedman 2012; Williams & Best 1982). Being intelligent is commonly associated with mathematical and spatial intelligence (Furnham et al. 2002). The brilliance = male stereotype has been shown to be endorsed even by children, both girls and boys, as young as 6-year-old (Bian et al. 2017; Cvencek et al. 2011), and it is believed to be a factor driving the underrepresentation of women in science, particularly in STEM fields (Nosek et al. 2009; Reuben et al. 2014; Smyth & Nosek 2015; Storage et al. 2016). An illustrative pattern is provided by Leslie et al. (2015), who show that women are underrepresented in academic fields where their practitioners believe that an innate talent (brilliance) is required to succeed. This evidence supports the hypothesis that the brilliance = male stereotype impacts the gender imbalance in STEM fields. In general, gender stereotypes in the workplace have been shown to hamper career aspirations of women (Smith et al. 2018), and an analogy can be made on gender stereotypes and formal studies.

Even when a consensus exists regarding the brilliance = male stereotype having strong cultural roots, studies on its perpetuation usually center on the analysis of cultural behaviors, such as differential guidance provided by parents to their offspring during shared scientific thinking (Crowley et al. 2001) or differential guidance given by science teachers to students according to their gender (Shumow & Schmidt 2013; Carlana 2018). Collecting further evidence on the presence of this type of stereotypes in cultural products remains as an open and active area of research.

Gender-Stereotyped Portrayals in Media

Researchers have been intensively studying gender-stereotyped portrayals in media for the last decades (for example, see Collins 2011; Koivula 1999; Silverstein et al. 1986). Even when the brilliance = male stereotype has not been the main focus of their studies, an extensive body of literature addresses how media portray female scientists and STEM professionals (Flicker 2003; Long et al. 2010; Orthia & Morgain 2016; Steinke 2005; Steinke & Tavarez 2018; Steinke et al. 2012). Previous literature has shown how the perception that children have of scientists is influenced by the way in which they are represented in media (Cheryan et al. 2013; Steinke et al. 2007; Tan et al. 2017). In particular, this literature suggests that, due to scarce regular contact of average children with STEM professionals, mass media has an important role in the dissemination of stereotypical images of scientists.

Additionally, studies report that this image usually involves a White man wearing a lab coat and glasses, exhibiting remarkable intelligence, and lacking interpersonal skills (Cheryan et al. 2013; Long et al. 2010; Steinke et al. 2007). This stereotype has been reported to clash with stereotypical feminine characteristics embedded in cultural products, and this clash is presumed to be detrimental to the self-identification of girls with scientific work. Note that self-identification is believed to be a relevant factor in the decision-making of girls when deciding to pursue a STEM career or not (Long et al. 2010; Steinke et al. 2012). Importantly, Cheryan et al. (2013) show that media interventions can be useful in fostering women's interest in STEM careers.

Social cognitive theory (Bandura et al. 1963) places the preceding findings into a theoretical framework. Concretely, it proposes that children acquire behavioral patterns and attitudes through continuous observation of symbolic models, such as the ones reproduced in

popular media. This theory explains how television and films reinforce and disseminate gender stereotypes through the way characters are portrayed. In this line, researchers have observed that media influence is intensified when children identify with the characters. For example, Maccoby and Wilson (1957) showed that children tend to identify more with same-gender characters and that they remember with more precision the actions and words of the characters with whom they identify. Furthermore, they observed that reinforcement in learning also depends on the actions performed by characters: Whereas violent content is easier to remember for boys, boy-girl interactive content is easier to remember for girls. Note that social cognitive theory highlights the importance of studying how women are portrayed in cultural products, for which the detection and characterization of these portrayals is a mandatory first step.

Culturomics

Culturomics, a term coined by Michel et al. (2011), refers to the study of large collections of cultural products (e.g., song lyrics, television shows, movies, books), and it stands as a promising approach for capturing general cultural viewpoints (Lamoreaux & Morling, 2012; Morling & Lamoreaux, 2008) as well as for examining cultural changes through time (Michel et al. 2011; Schwartz et al. 2013; Twenge et al. 2012). Given that culture serves as a source for acquiring individual values and behaviors (a process known as enculturation), biases and stereotypes embedded in these products are regarded as relevant factors affecting this acquisition (Cape 2003; Crowley et al. 2001; Tenenbaum & Leaper 2003). For this reason, several recent studies analyze and quantify the presence of a wide range of biases and stereotypes in large collections of cultural products (Dimnik & Felton 2006; Garg et al. 2018; Ramakrishna et al. 2017; Twenge et al. 2012). The present study follows this line of research.

The Present Study

We make use of modern computational methods to study the presence of the brilliance = male stereotype in a large collection of movie transcripts covering half a century of Western-world film history. For this, we gathered a large-corpus of English subtitles of Western-world high-grossing films released since 1967 up to and including 2016 ($n = 11,550$). Because stereotypes are, in part, a collection of associations that link a group to a set of descriptive characteristics (Gaertner & McLaughlin 1983), we used natural language processing techniques to quantify associations between gender and high-level cognitive ability in films during the last 50 years.

Concretely, we measured word associations by computing positive pointwise mutual information (PPMI) scores (Church & Hanks 1990; Jurafsky & Martin 2000) between gender pronouns (female pronouns: she, hers, her, herself; male pronouns: he, his, him, himself) and high-level cognitive ability-related words (e.g., genius, intelligent, clever). PPMI is a metric designed to capture associations between words/concepts by measuring if they typically appear nearby each other (i.e., higher values meaning higher associations). Because our corpus allows us to estimate PPMI values, taking as input films of particular genres, we also analyze the presence of this stereotype in films belonging to the family and/or animation genres ($n = 1,041$). Note that because studies have shown that children can incorporate and reproduce gender stereotypes at an early age and that children learn about gender roles and stereotyped behavior through exposure to cultural definitions of gender (Bem 1981; Steinke 2005), movies belonging to family and animated genres, which are commonly targeted at children, are expected to impact in a more direct way.

Method

Data Collection

With the aim of identifying popular films over a period covering half a century, we downloaded lists of the 1,000 top-grossing U.S. titles for every year since 1967 up to and including 2016 from IMDb (www.imdb.com). These data included an internal IMDb id. Using this id number and the IMDbPY library (imdbpy.sourceforge.io), we downloaded metadata on each popular title. Metadata includes details on its kind (movie, television series, television movie, etc.), year of release, language(s) (English, French, Spanish, etc.), genre(s) (adventure, comedy, family, animation, etc., where one movie may belong to more than one genre), and country(ies) involved in its production. Given that our analysis requires films under study to share their language, we opted to analyze English-spoken movies (because these films have the largest box office gross revenue in the world). For this analysis, we filtered out all titles which were not movies, kept films which included English among the languages spoken in them, and kept films in which the United States, United kingdom, Canada, or Australia were involved in their production. This left us with a set of Western-world high-grossing films.

Then, for each movie in the resulting set, if available, we obtained from OpenSubtitles (www.opensubtitles.org) its most frequently downloaded English subtitle. Subtitles were obtained as SubRip files. A SubRip file contains groups/frames of formatted text lines separated by blank lines, each group/frame being numbered sequentially and having time-stamps indicating when it should appear on-screen. This process left us with our sample of 11,550 film subtitles spanning half a century of film history. It is important to remark that not all movies had subtitles available during data collection (particularly older ones did not).

Table 1 details how genre participation evolved as time passed by, as well as how it is distributed in the whole corpus. Each cell indicates the number of films that belong to a given genre for a particular time period. In broad terms, participation remained quite stable during the whole period for most genres, but some fluctuations can also be seen. For example, in some genres a moderate rise is observed in their participation (e.g., animation, family, romance, thriller), in others a moderate fall is observed (e.g., musical, war, western), and in others a sharp rise in particular time periods is observed (e.g., horror in the time period 1977-1986).

Figure 1A details the number of films analyzed for successive 10-year periods (1967-1976, 1977-1986, ..., 2007-2016) for the full sample and the family/animation subsample. Note that both samples grow considerable in size for years closer to the present and that the family/animation subsample represents a small fraction of the full sample. Additionally, Figure 1B shows the evolution of the ratio between the number of appearances of male pronouns relative to the number of appearances of female pronouns. In line with previous research on books (Twenge et al. 2012), in films, this ratio has experienced a reduction since the mid-1960s—a phenomenon believed to be associated with an improvement in women's status (Twenge et al. 2012)—and has been consistently less favorable toward female pronouns in the family/animation subsample when compared to the full sample. Note that this ratio is also consistent with results obtained by the Geena Davis Institute on Gender in Media (2015) when analyzing screen and speaking time by gender in a smaller set of movies.

Co-Occurrence Matrix

In the present work we analyzed word associations through PPMI estimates and evaluated their effect size and statistical significance using odds ratios. Both of these measures take as input co-occurrence matrices (C). A co-occurrence matrix presents the number of times a

token appears in the context of another one (each row representing a target token and each column representing a context token). By construction, these matrices are square and symmetric (of size $|V| \times |V|$, being V the vocabulary). In this work, context is defined as all subtitles' frames which appear close in time to the token under analysis. This depends on the size of a time window, which has to be pre-defined (we set this value equal to 30 seconds in this work). Figure 2 contains a snippet illustrating how, given a set of subtitles, these matrices are built. An interesting technical feature of our analysis is that, given that subtitles contain time information, we were able to modify co-occurrence matrix construction algorithms in order to exploit the time dimension. Note that one can build co-occurrence matrices for different subsamples of films (e.g., films from 2010 onwards, family/animation films) and consequently analyze the presence of associations of interest in particular subsets of films.

Positive Pointwise Mutual Information

Pointwise mutual information (PMI) is a measure of association which captures if two tokens co-occur more/less often than chance (Church & Hanks 1990; Jurafsky & Martin 2000). In this way, it captures what is known as first-order co-occurrences (sometimes called syntagmatic associations; Schütze & Pedersen 1993). It is defined as:

$$PMI(w, c) = \log_2 \frac{P(w, c)}{P(w)P(c)} \quad (1)$$

where w stands for the target token and c stands for the context token. Note that the numerator indicates how often two tokens are observed together, whereas the denominator indicates how often they are expected to co-occur assuming independence. It should also be noted that, although third words in the contexts may influence the way a given pair of words are associated (e.g., in the case of negations), in studies such as ours (where there is extensive data), it is

assumed that these effects are not the norm and that they are evenly distributed across phrases, not affecting the general patterns.

Given that negative PMI values tend to be unreliable (except for enormous corpora; Jurafsky & Martin 2000), we opted to measure associations using PPMI estimates. PPMI is defined as:

$$PPMI(w, c) = \max(\log_2 \frac{P(w, c)}{P(w)P(c)}, 0) \quad (2)$$

All probabilities contained in Equation 2 can be estimated from a co-occurrence matrix. Instead of using the traditional formulas presented in Jurafsky & Martin (2000), and given that we estimate associations between sets of words, we estimate these probabilities according to the following equations:

$$P(w, c) = \frac{\sum_{i \in w} \sum_{j \in c} C_{i,j}}{\sum_{i=1}^{|V|} \sum_{j=1}^{|V|} C_{i,j}} \quad (3)$$

$$P(w) = \frac{\sum_{i \in w} \sum_{j=1}^{|V|} C_{i,j}}{\sum_{i=1}^{|V|} \sum_{j=1}^{|V|} C_{i,j}} \quad (4)$$

$$P(c) = \frac{\sum_{i=1}^{|V|} \sum_{j \in c} C_{i,j}^\alpha}{\sum_{i=1}^{|V|} \sum_{j=1}^{|V|} C_{i,j}^\alpha} \quad (5)$$

where, in our case, w is a set containing the row indices of the target tokens of interest (i.e., female or male gender pronouns), c is a set containing the columns indices of the context tokens with which the association is going to be measured (e.g., feminine stereotyping roles; high-level cognitive ability-related words), and α is a parameter found to improve the performance by reducing the bias toward low frequency events; as suggested in the literature (Jurafsky & Martin, 2000), we set its value equal to 0.75.

Effect Sizes and Tests of Statistical Significance

Given a co-occurrence matrix (C) and three sets of tokens w_1 (e.g., female pronouns), w_2 (e.g., male pronouns), and c (e.g., high-level cognitive ability-related words), contingency tables can be used to analyze effect sizes and statistical significance of words' associations. To do so, one needs to first construct a 2×2 contingency table (CT) by first counting how many times tokens in c were said in the contexts of tokens included in w_1 ($C_{1,1} = \sum_{i \in w_1} \sum_{j \in c} C_{i,j}$) and in the contexts of tokens included in w_2 ($C_{2,1} = \sum_{i \in w_2} \sum_{j \in c} C_{i,j}$). Then, in a similar fashion, one has to count how many times tokens in $\neg c$ —all words in the vocabulary not included in c —were said in the contexts of tokens included in w_1 ($C_{1,2} = \sum_{i \in w_1} \sum_{j \notin c} C_{i,j}$) and in the contexts of tokens included in w_2 ($C_{2,2} = \sum_{i \in w_2} \sum_{j \notin c} C_{i,j}$). Taking these 2×2 contingency tables as input, one can estimate the odds of tokens in c being said in the contexts of tokens in w_1 relative to tokens in $\neg c$ being said in the context of tokens in w_1 ($Odds_1 = CT_{1,1}/CT_{1,2}$), as well as the odds of tokens in c being said in the contexts of tokens in w_2 relative to tokens in $\neg c$ being said in the context of tokens in w_2 ($Odds_2 = CT_{2,1}/CT_{2,2}$).

Finally, taking these odds as inputs, one can estimate an odds ratio ($OR = Odds_1/Odds_2$). Odds ratios are effect size measures (Haddock et al. 1998; Olivier & Bell 2013). Values larger than 1 indicate that the association between w_1 and c is relatively stronger than the one observed between w_2 and c , whereas the opposite can be inferred from values smaller than 1. As a general rule, odds ratios greater (lesser) than 1.68 (0.60), 3.47 (0.29), and 6.71 (0.15) suggest small (but not negligible), medium, and large effects (Chen et al. 2010). An additional advantage of odds ratios is that, by taking the natural logarithm of their value, a statistical significance test can be carried out (Haddock et al. 1998). The test's null hypothesis is that the natural logarithm of the odds ratio equals 0 (i.e., that the odds ratio equals 1).

Sets of Words Used

Here we provide details on the words included in the different sets of words considered in the analysis. As in Twenge et al. (2012), we used gender pronouns to check on gender discrepancies. The set of *female pronouns* includes the following words: she, hers, her, and herself. The set of *male pronouns* includes the following words: he, his, him, and himself.

To check that our proposed methodology effectively captures patterns already reported in the literature, we will first study associations between gender pronouns and sets of words considered to reflect stereotyping and non-stereotyping roles. These sets of words were obtained from Lenton et al. (2009). There, the authors report that 143 online participants rated 154 words with respect to their masculinity/femininity (1 = very masculine; 7 = very feminine) and their valence (1 = very negative; 7 = very positive). The authors also report that this was done with the aim of obtaining sets of masculine, feminine, and gender-neutral words that were matched in several respects (valence, word length, word frequency) other than their gendered nature.

The set of *feminine stereotyping roles* includes the following words: beautician, caregiver, cheerleader, dancer, decorator, designer, dietician, florist, hairdresser, homemaker, housekeeper, model, nanny, nurse, receptionist, stylist, and typist. The set of *masculine stereotyping roles* includes the following words: architect, carpenter, coach, contractor, detective, electrician, engineer, farmer, firefighter, gambler, inventor, machinist, mechanic, officer, physicist, pilot, programmer, rancher, sheriff, and soldier. The set of *non-stereotyping roles* includes the following words: assistant, cashier, clerk, doctor, editor, lawyer, poet, reporter, servant, and worker.

We were not able to obtain a well-established list of high-level cognitive ability-related words from the literature. For this reason we opted to build our own. To do so, we followed the

following steps. From Bian et al. (2017) we identified two nouns associated to the brilliance = male stereotype: brilliance and genius. We instantiated our set of words using these two nouns. We omitted giftedness from Bian et al. (2017) because its main definition relates to the natural ability to do something well, not to intelligence. Using the Collins Thesaurus (www.collinsdictionary.com) we added cleverness to the set because it is considered a synonym of brilliance. Then, we added intelligence to the set, because it is considered a synonym of cleverness by the same source. We also added brightness and ingeniousness to the set for being considered interchangeable words with brilliance and genius, respectively. Finally, to further expand the set of high-level cognitive ability-related words, we added the adjective and adverb forms of these nouns. The process resulted in the following set of *high-level cognitive ability-related words*: ingenious, genius, ingeniousness, ingeniously, bright, brightness, brightly, brilliant, brilliance, brilliantly, clever, cleverness, cleverly, intelligent, intelligence, and intelligently.

To check for robustness in our results, we also analyzed associations between gender pronouns and an alternative list of high-level cognitive ability-related words obtained from Garg et al. (2018) (“intellectual words” in the pre-print version; “competence adjectives” in the final manuscript). This alternative set consists of the following words: adaptable, analytical, apt, astute, brilliant, clever, discerning, genius, imaginative, ingenious, inquiring, inquisitive, intelligent, intuitive, inventive, judicious, logical, luminous, precocious, reflective, resourceful, sagacious, sage, shrewd, smart, thoughtful, venerable, and wise.

Results

As we already mentioned, before focusing on the presence of the brilliance = male stereotype, and as a sanity check, we explored if our framework effectively captures the presence

of gender stereotyping roles found when analyzing large collection of textbooks, novels and newspapers. Concretely, as previously detailed, from Lenton et al. (2009) we retrieved carefully constructed and curated lists of feminine stereotyping roles, masculine stereotyping roles, and non-stereotyping roles, and we then analyzed its associations with female and male pronouns.

Figure 3A presents PPMI estimates considering films from 2010 up to and including 2016 for the full sample and the family/animation subsample. Estimates for the full sample indicated that our framework effectively captures stereotyping roles associations. First, female pronouns had a greater association with feminine stereotyping roles (PPMI = 2.71) than male pronouns did (PPMI = 1.99), and the effect size of the difference according to the estimated odds ratio was equal to 1.64 and was statistically significant ($p < .001$). Second, female pronouns had a smaller association with masculine stereotyping roles (PPMI = 1.88) than male pronouns did (PPMI = 2.67), and the effect size of the difference according to the estimated odds ratio was equal to 0.58 and was statistically significant ($p < .001$). Third, female and masculine pronouns had an almost similar association with non-stereotyping roles (PPMI = 2.64 and PPMI = 2.62 respectively), and the effect size of the difference according to the estimated odds ratio was equal to 1.01 and was not statistically significant ($p = .365$).

Notably, our estimates suggest that the strength of these gender-stereotyping associations was greater in films targeted at children. Once again female pronouns had a greater association with feminine stereotyping roles (PPMI = 2.97) than male pronouns did (PPMI = 1.49), and the effect size of the difference according to the estimated odds ratio was equal to 2.79 and was statistically significant ($p < .001$). Female pronouns had a smaller association with masculine stereotyping roles (PPMI = 1.53) than male pronouns did (PPMI = 2.67), and the effect size of the difference according to the estimated odds ratio was equal to 0.45 and was statistically

significant ($p < .001$). Female and masculine pronouns had a similar association with non-stereotyping roles (PPMI = 2.53 and PPMI = 2.57 respectively), and the effect size of the difference according to the estimated odds ratio was equal to 0.98 and was not statistically significant ($p < .699$).

Figure 3B explores the dynamics of these associations taking as input sets of films belonging to successive 10-year periods. Full-sample estimates revealed that stereotyping associations showed far from converging trends during the last 50 years. Estimates based on the family/animation subsample revealed a qualitatively similar picture, although they were much less stable (we attribute this to the fact that sample sizes for every 10-year period of the family/animation subsample were much smaller than their full sample counterparts) (see Table 1 and Figure 1A).

We now place our focus on quantifying the presence of the brilliance = male stereotype in films by studying associations between gender pronouns and words depicting high-level cognitive ability. Figure 4A presents estimates considering all movies from 2010 up to and including 2016 for the full sample and the family/animation subsample. Estimates indicate that the association between female pronouns and high-level cognitive ability-related words (PPMI = 1.94) was smaller than the one male pronouns had with high-level cognitive ability-related words (PPMI = 2.27), where the effect size of the difference according to the estimated odds ratio was equal to 0.79 and was statistically significant ($p < .001$). When the focus is placed in the family/animation subsample, estimates again indicated that the association between female pronouns and high-level cognitive ability-related words (PPMI = 1.77) was smaller than the one male pronouns had with high-level cognitive ability-related words (PPMI = 2.17), where the

effect size of the difference according to the estimated odds ratio was equal to 0.76 and was statistically significant ($p < .001$).

Figure 4B explores the dynamics of these differences through time. Results from the full sample of movies revealed that the difference has been steady at least for half a century, and, once again, results showed no evidence of convergence in the trends. Results from the family/animation subsample revealed a qualitatively similar, but less stable, pattern. Only for the 1967-1976 period, estimates indicated that high-level cognitive ability-related words had a greater association with female pronouns than with male pronouns, although a statistical significance test based on the log odds ratio did not reveal this difference to be statistically significant ($p = .405$).

To check for robustness in our results, Figure 5 replicates Figure 4's analysis but instead of using our set of high-level cognitive ability-related words, we used the set of competence adjectives presented in Garg et al. (2018) (intellectual words in the pre-print version). Results remained robust. In particular, the full-sample estimates indicated that the association between female pronouns and high-level cognitive ability-related words (PPMI = 2.32) was smaller than the one between male pronouns and high-level cognitive ability-related words (PPMI = 2.46), with the effect size of the difference according to the estimated odds ratio equal to 0.90 and being statistically significant ($p < .001$). When the focus is placed in the family/animation subsample, estimates again indicated that the association between female pronouns and high-level cognitive ability-related words (PPMI = 1.84) was smaller than the one between male pronouns and high-level cognitive ability-related words (PPMI = 2.39), with an effect size of the difference according to the estimated odds ratio equal to 0.69 that is statistically significant ($p < .001$).

Discussion

Overall, our estimates suggest that stereotyping associations are present in films. In line with previous literature, we effectively find stereotyping associations for the case of stereotyping roles (which serves as a validation of our methodology). When the focus is placed in the brilliance = male stereotype, our results suggest that it is also prevalent in films. Nevertheless, it should be noted that the estimated effect size of these associations were smaller than the ones found for stereotyping roles (although they pass statistical significance tests). Moreover, the reported patterns seem to have been quite persistent for the last 50 years and have shown little evidence of convergence. Notably, when the focus is placed on movies commonly aimed at children, all these patterns remain. In fact, effect size estimates suggest that stereotyping associations are as large or even larger in the family/animation subsample than they are in the full sample.

Limitations

The computational framework presented in our study, by analyzing films' content through the analysis of subtitles, only captures the presence of gender biases embedded in spoken dialogue. A drawback of this approach is that it leaves aside other dimensions of films which may contain stereotypical associations, such as films' images and non-verbal cues in audio. (Note that this point is addressed by the Geena Davis Institute on Gender in Media 2015.) Analyzing images and audio at a large scale could open the door to the study of other stereotyping elements, such as clothing, actions, gestures, and voice prosody. Future research could move forward in these directions.

Additionally, our results depend to a great extent on the words used as input in the analysis. Extra caution should be taken when choosing the words to include because they

determine which stereotype is being quantified and which hypothesis is being tested. This highlights the fact that when studying gender associations, the role of specialists and specialized literature cannot be underestimated.

Future Research Directions

Empirical evidence shows that scientific careers tend to be associated with women in some non-Western countries (Larivière et al. 2013; Stoet & Geary 2018). An interesting research line is the analysis of how these stereotypical associations vary between film industries around the world. Additionally, we only focus on analyzing gender stereotypes related to intelligence/brilliance. However, the proposed methodology can be used to study other stereotypes that are related to gender or not. Given its potential for being useful in other research questions, we plan to make our framework available in the near future to the research community as a free online service.

Finally, the framework we use in the present study places its focus in studying words' associations in large corpora. Adapting these methods to single documents poses several methodological challenges because they require large volumes of data to work properly (something a single movie does not provide). Future research could focus in developing techniques for identifying and ranking stereotyping associations at the film level.

Practice Implications

Nowadays, gender biases are being strongly discussed in society in general, especially in relation to the Western-world film industry. Controversies range from the existence of a strong gender pay gap (actors being paid considerably more than actresses) up to allegations of widespread prevalence of sexual assault and harassment. Our findings indicate that gender inequality is also considerably strong in the contents of films. We believe that both the evidence

we provide and the methods we developed contribute to this debate. Moreover, by automatically detecting biases present in cultural products which we consume everyday, the framework presented in our article can be useful for activists, policymakers, and the scientific community concerned in studying and reducing stereotyping associations in mass media, whether gender-related or not.

Conclusion

In our paper we applied data science and computational linguistics techniques to quantify the presence of the brilliance = male stereotype in a large collection of films' transcripts covering half a century of Western-world film history. It should be noted that, even when there is a collection of previous research that uses traditional methods to analyze gender portrayals in cultural products, as far as we know, ours is the first study to analyze stereotypes in such a large corpus of films.

Our results indicate that the brilliance = male stereotype is significantly prevalent in films. Moreover, a temporal analysis of this phenomenon shows that this has remained quite constant for the last 50 years. Importantly, when a subset of films belonging to genres particularly targeted at children (i.e., family and/or animation) is analyzed, and the same pattern of stereotypical associations is observed. Our findings indicate that gender inequality is considerably strong in the contents of Western-world films. Given that stereotypes have been found to shape intellectual identity and performance (Steele 1997), the need to proactively address the presence of the brilliance = male stereotype in films is evident.

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Table 1

Distribution of films' genres through time.

Genre	Time period					
	1967-1976 <i>n</i> (%)	1977-1986 <i>n</i> (%)	1987-1996 <i>n</i> (%)	1997-2006 <i>n</i> (%)	2007-2016 <i>n</i> (%)	1967-2016 <i>n</i> (%)
Action	183 (16.68%)	240 (18.18%)	416 (20.16%)	475 (16.36%)	684 (16.41%)	1,998 (17.30%)
Adventure	141 (12.85%)	238 (18.03%)	243 (11.78%)	372 (12.81%)	517 (12.41%)	1,511 (13.08%)
Animation	12 (1.09%)	38 (2.88%)	50 (2.42%)	112 (3.86%)	165 (3.96%)	377 (3.26%)
Biography	48 (4.38%)	56 (4.24%)	102 (4.94%)	159 (5.48%)	328 (7.87%)	693 (6.00%)
Comedy	327 (29.81%)	459 (34.77%)	861 (41.74%)	1,238 (42.65%)	1,380 (33.12%)	4,265 (36.93%)
Crime	215 (19.60%)	193 (14.62%)	448 (21.72%)	548 (18.88%)	615 (14.76%)	2,019 (17.48%)
Documentary	29 (2.64%)	39 (2.95%)	40 (1.94%)	138 (4.75%)	405 (9.72%)	651 (5.64%)
Drama	571 (52.05%)	599 (45.38%)	1,090 (52.84%)	1,622 (55.87%)	2,220 (53.28%)	6,102 (52.83%)
Family	70 (6.38%)	82 (6.21%)	200 (9.69%)	273 (9.40%)	337 (8.09%)	962 (8.33%)
Fantasy	56 (5.10%)	106 (8.03%)	214 (10.37%)	246 (8.47%)	361 (8.66%)	983 (8.51%)
History	42 (3.83%)	35 (2.65%)	58 (2.81%)	101 (3.48%)	177 (4.25%)	413 (3.58%)
Horror	153 (13.95%)	253 (19.17%)	274 (13.28%)	274 (9.44%)	553 (13.27%)	1,507 (13.05%)
Music	36 (3.28%)	70 (5.30%)	82 (3.97%)	144 (4.96%)	197 (4.73%)	529 (4.58%)
Musical	54 (4.92%)	49 (3.71%)	46 (2.23%)	50 (1.72%)	63 (1.51%)	262 (2.27%)
Mystery	94 (8.57%)	116 (8.79%)	161 (7.80%)	309 (10.64%)	398 (9.55%)	1,078 (9.33%)
News	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	18 (0.43%)	18 (0.16%)
Romance	162 (14.77%)	227 (17.20%)	471 (22.83%)	794 (27.35%)	829 (19.89%)	2,483 (21.50%)
Sci-Fi	89 (8.11%)	184 (13.94%)	193 (9.36%)	254 (8.75%)	370 (8.88%)	1,090 (9.44%)
Short	3 (0.27%)	13 (0.98%)	15 (0.73%)	38 (1.31%)	21 (0.50%)	90 (0.78%)
Sport	19 (1.73%)	41 (3.11%)	66 (3.20%)	118 (4.06%)	138 (3.31%)	382 (3.31%)
Thriller	245 (22.33%)	317 (24.02%)	587 (28.45%)	778 (26.80%)	1,180 (28.32%)	3,107 (26.90%)
War	69 (6.29%)	60 (4.55%)	55 (2.67%)	106 (3.65%)	128 (3.07%)	418 (3.62%)
Western	118 (10.76%)	32 (2.42%)	39 (1.89%)	33 (1.14%)	39 (0.94%)	261 (2.26%)
# of films	<i>n</i> = 1,097	<i>n</i> = 1,320	<i>n</i> = 2,063	<i>n</i> = 2,903	<i>n</i> = 4,167	<i>n</i> = 11,550

Note. Each cell indicates the number of films which belong to a given genre for a particular time period. As most movies are catalogued as belonging to more than one genre, columns do not add exactly up to 100%.

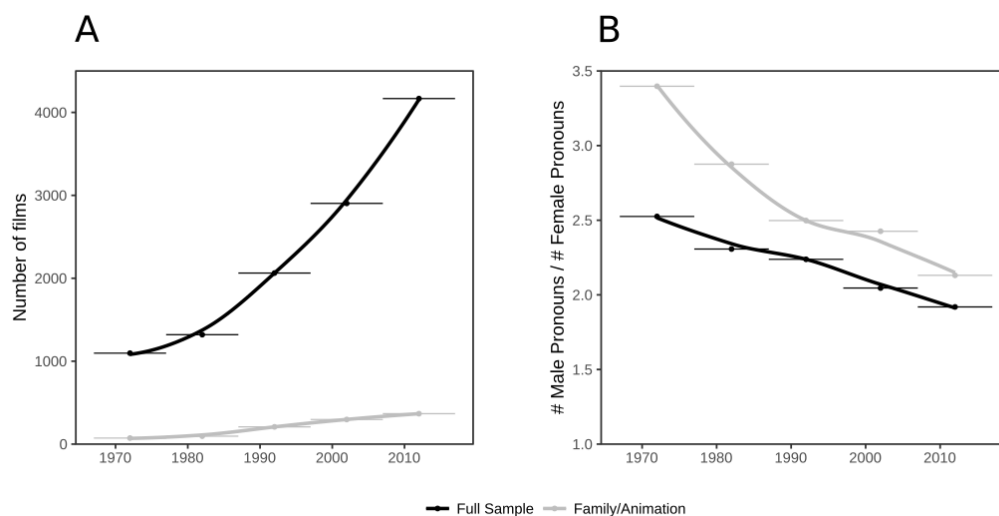


Figure 1. Film frequencies and gender pronouns ratios for successive 10-year periods. (A) Number of films analyzed for successive 10-year periods (1967-1976, 1977-1986, ..., 2007-2016), for the full sample and the family/animation subsample. (B) Evolution of the ratio between the number of appearances of male pronouns (he, his, him, himself) relative to the number of appearances of female pronouns (she, hers, her, herself) for the full sample and the family/animation subsample. In both panels, tendencies are estimated through LOESS regressions.

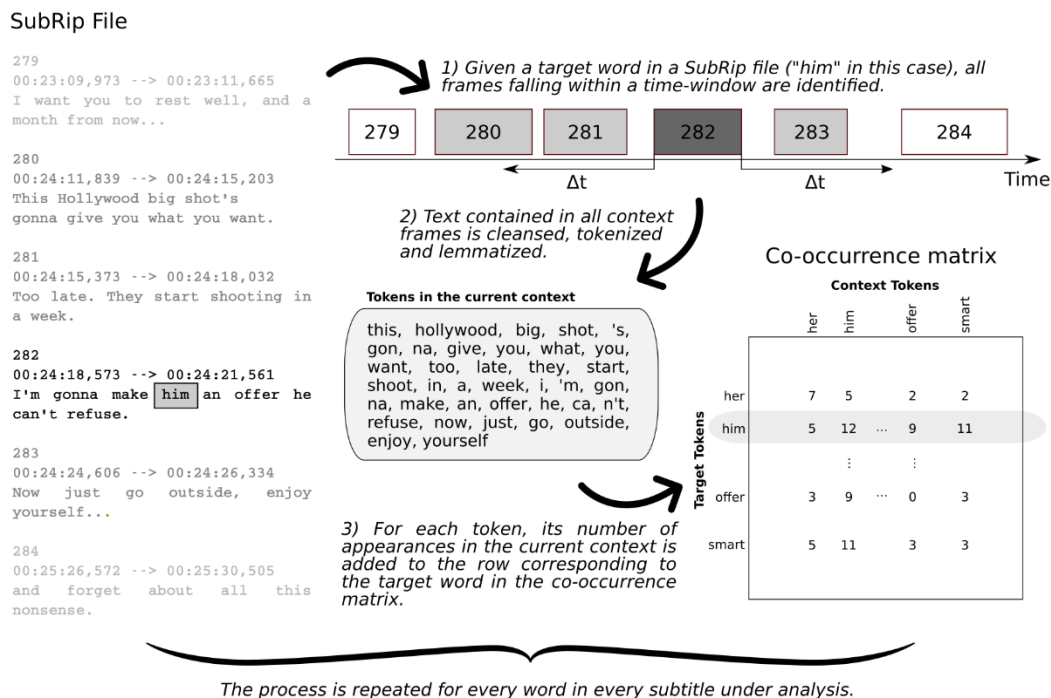


Figure 2. Co-occurrence matrix construction. Given a target word in a SubRip file ("him" in the illustration), all neighboring/context frames are identified. Which frames constitute the neighborhood depends on the size of a time window (Δt), which we set equal to 30 seconds in this work. The text contained in all context frames is cleansed, tokenized, and lemmatized. Then, the number of appearances of every context token is added to the relevant cell in the co-occurrence matrix under construction. The process is repeated for every word in every subtitle under analysis. A co-occurrence matrix presents the number of times a word appears in the contexts of another word (for example, and simply as an illustration, according to this figure smart appears eleven times in the contexts of him), and it serves as input for PPMI estimates, odds ratios estimates, and statistical significance tests.

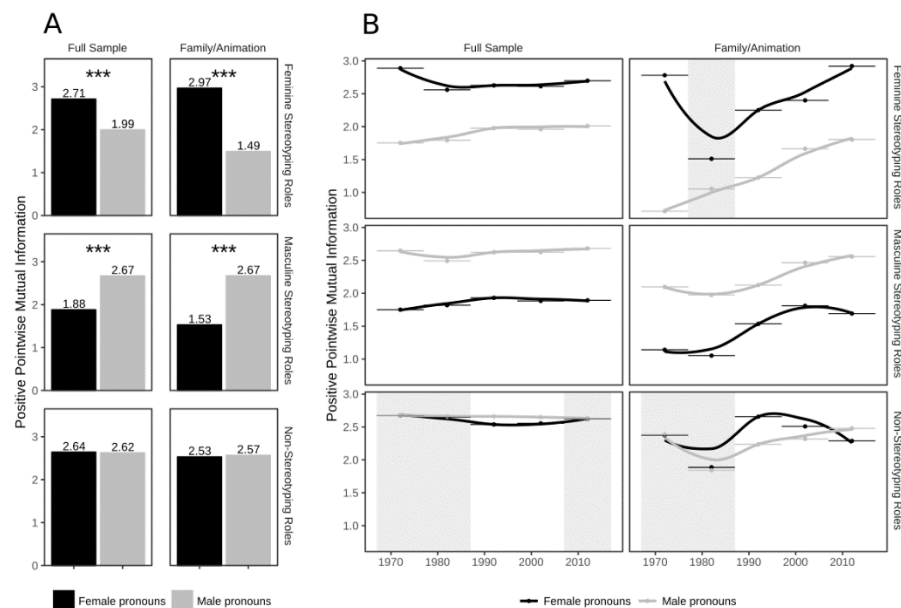


Figure 3. Word associations between gender pronouns and stereotyping roles. (A) Estimated associations between gender pronouns and stereotyping roles when analyzing films from 2010 up to and including 2016 for the full sample ($n = 2,902$) and the family/animation subsample ($n = 242$). Asterisks indicate the results of log odds ratios statistical significance tests on the underlying contingency tables. (B) Time evolution of the estimated associations taking as input sets of films belonging to successive 10-year periods (1967-1976, 1977-1986, ..., 2007-2016). Tendencies are estimated through LOESS regressions. Gray areas indicate that, according to log odds ratios statistical significance tests, differences are not significant at the 5% level.

*** $p < .001$.

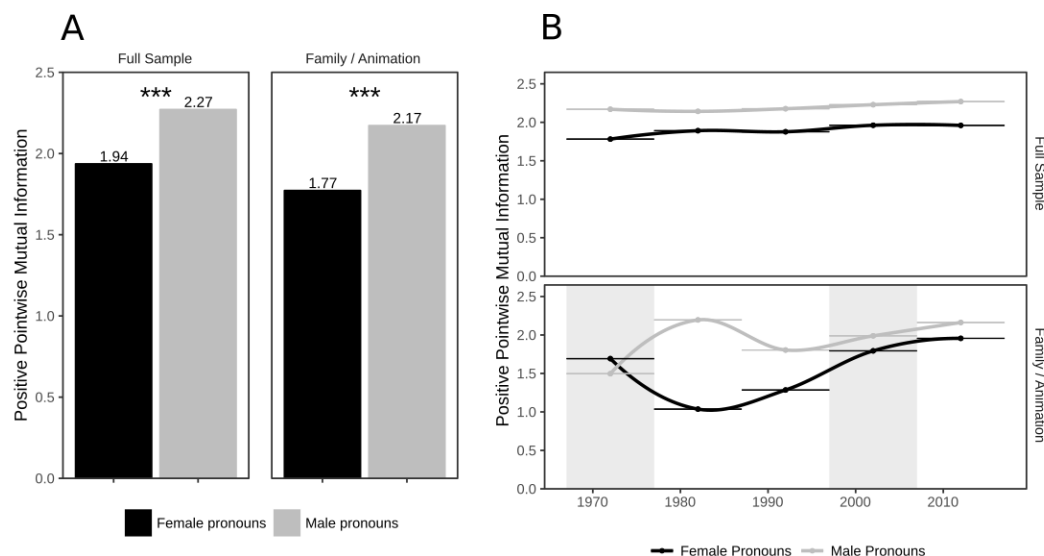


Figure 4. Word associations between gender pronouns and high-level cognitive ability-related words. (A) Estimated association between gender pronouns and high-level cognitive ability-related words when films from 2010 up to and including 2016 are analyzed for the full sample ($n = 2,902$) and the family/animation subsample ($n = 242$). Asterisks indicate the results of log odds ratios statistical significance tests on the underlying contingency tables. (B) Time evolution of the estimated associations taking as input sets of films belonging to successive 10-year periods (1967-1976, 1977-1986, ..., 2007-2016). Tendencies are estimated through LOESS regressions. Gray areas indicate that, according to log odds ratios statistical significance tests, differences are not significant at the 5% level.

*** $p < .001$.

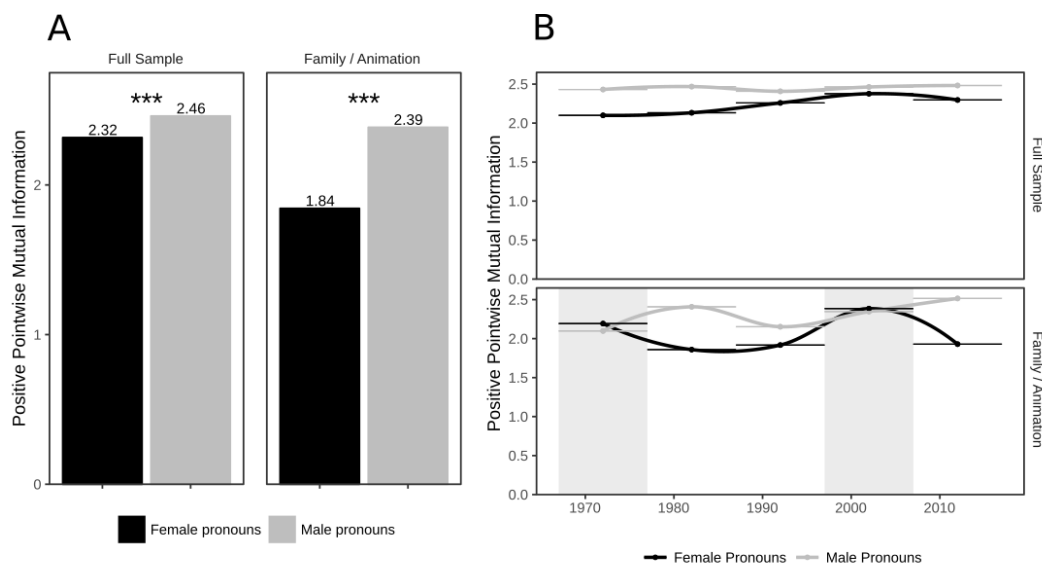


Figure 5. Word associations between gender pronouns and competence adjectives. (A) Estimated association between gender pronouns and competence adjectives (set of words taken from Garg et al., 2018) when films from 2010 up to and including 2016 are analyzed for the full sample ($n = 2,902$) and the family/animation subsample ($n = 242$). Asterisks indicate the results of log odds ratios statistical significance tests on the underlying contingency tables. (B) Time evolution of the estimated associations taking as input sets of films belonging to successive ten-year periods (1967-1976, 1977-1986, ..., 2007-2016). Tendencies are estimated through LOESS regressions. Gray areas indicate that, according to log odds ratios statistical significance tests, differences are not significant at the 5% level.

*** $p < .001$.